ECON 0100 | Exercise C2 | Taxes & Subsidies

In recent years Hogsmeade has seen a proliferation of candy shops selling smoked toffees. The Demand (marginal benefit) curve and Supply (marginal cost) curve for smoked toffee can be represented by the following relationships.

$$D : P_B = 20 - \frac{1}{2}Q_D \qquad\qquad S : P_S = 2 + Q_S$$

Due to the metabolic issues among the younger generation of wizards in the community resulting from the consumption of these high glycemic index treats, the Ministry of Magic imposed a 3 Galleon tax on the sale of smoked toffee.

Q1 | One Toffee Exchange

First Exchange: A buyer values one smoked toffee at 16 Galleons. It costs the seller 12 Galleons to make.

a) Without the tax, will the buyer and seller make this exchange? _____

b) With the tax, will the buyer and seller make this exchange? _____

c) Total Surplus to the buyer and seller, with the tax: _____

d) Tax paid to the Ministry of Magic: _____

e) Total Surplus, including the Ministry of Magic: _____

Another buyer values one smoked toffee at 14 Galleons. It costs the seller 12 Galleons to make.

f) Without the tax, will the buyer and seller make this exchange? _____

g) With the tax, will the buyer and seller make this exchange? _____

h) Deadweight Loss from losing this exchange: _____

Q2 | Tax Equilibrium

Use a graph and algebra to discuss the impact of the tax on the market. Include how the welfare areas changed after the policy. No need to calculate welfare measures except DWL.

Pre-Tax Price: _____ Pre-Tax Quantity: _____

Post-Tax Quantity: _____ DWL: _____

Buyer Price: _____ Seller Price: _____

Tax Incidence on Buyers: _____ Tax Incidence on Sellers: _____

Who bears more of the tax? *buyers* · *sellers* · *equal*

Q3 | Subsidy Equilibrium

Now suppose that instead of the tax, the Ministry of Magic pays a 3 Galleon subsidy on every smoked toffee sold. Use a graph and algebra to discuss the impact of the subsidy on the market.

Post-Subsidy Quantity: _____ DWL: _____

Buyer Price: _____ Seller Price: _____